

DICE COUNTERFACTUAL EXPLANATIONS

"What-If" Analysis for Loan Decisions

Generated: 2026-04-16 18:57:31

WHAT IS DiCE (Diverse Counterfactual Explanations)?

Counterfactual explanations answer:

"What would need to change for a DIFFERENT prediction?"

Example:

- Applicant John:
 - Current prediction: 25% default risk (APPROVED)
 - Scenario 1: IF credit score dropped 50pts → 35% risk
 - Scenario 2: IF debt-to-income rose 10% → 40% risk
 - Scenario 3: IF all optimal → 10% risk (DISAPPROVED)

Business Value:

- ✓ Show applicants what would improve their approval
- ✓ Negotiate with applicants on realistic changes
- ✓ Identify most impactful factors for THIS applicant
- ✓ More actionable than global SHAP explanations

KEY DIFFERENCES FROM SHAP/LIME:

SHAP/LIME: "WHY did they get that score?"

- Feature importance → How features affected average
- Good for understanding model behavior

DiCE: "WHAT would change the decision?"

- Actionable counterfactuals → Why specific applicant didn't qualify
- Good for customer service and appeals

USE CASES:

- ✓ Appeal process: Show applicants path to approval
- ✓ Loan officer guidance: Recommend actions to improve change
- ✓ Risk management: Identify critical thresholds per applicant
- ✓ Customer communication: Explain what would help
- x Automated decision making (too specific/unstable)

INTERPRETATION GUIDE:

Each counterfactual scenario shows:

1. ORIGINAL values → applicant's current situation
2. COUNTERFACTUAL values → "what-if" changes
3. IMPACT → how much risk changes

Features marked as "FEASIBLE" = realistic changes

Features marked as "COSTLY" = difficult to change

Data Used: Original (Non-WoE) Feature Scale

Easily understandable by loan officers and customers

IMPLEMENTATION GUIDE
Using DiCE Explanations in Production

DEPLOYMENT CHECKLIST:
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- 1. CUSTOMER COMMUNICATION
How to share with declined applicants:

Email Template:
"Thank you for your application. Based on our analysis, we recommend focusing on these factors that would most improve your approval odds:
 - Reduce monthly debt obligations by \$X
 - Build credit history for 6 months
 - Save for 5-10% additional down payment
We'd be happy to revisit in 6 months!"

- 2. LOAN OFFICER INTEGRATION
Train team on reading counterfactual tables:
 - Original = what applicant currently has
 - Counterfactual = what would make them approvable
 - Feasible = realistic in 6-12 months
Use in appeals: "Here's what would have changed the decision"

- 3. RISK COMMITTEE REPORTING
Use DiCE to understand portfolio thresholds:
 - How many applicants 1-2 factors away from approval?
 - What's the distribution of "needed improvements"?
 - Are there market segments we're missing?

- 4. PRODUCT DESIGN
Design products around counterfactuals:
 - Credit builder loans for those needing credit history
 - Down payment assistance for those short on equity
 - Co-signer programs for income below threshold

INTERPRETATION RULES:
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- When features show significant changes:
- DON'T assume applicant can/should make all changes
 - ✓ DO prioritize top 2-3 factors
 - ✓ DO verify feasibility before sharing with customer
 - ✓ DO follow up in 6 months
- When counterfactuals seem unrealistic:
- This indicates applicant is far from approval threshold
 - Flag for committee review of pricing/risk assessment
 - Consider portfolio risks if many similar applicants

MONITORING & IMPROVEMENT:
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- Monthly:
- ✓ Track how many declined applicants reapply
 - ✓ Monitor those who made recommended changes
 - ✓ Update counterfactual scenarios with new data
- Quarterly:
- ✓ Review which features most commonly appear in scenarios
 - ✓ Adjust product offerings based on patterns
 - ✓ Retrain model with new applicant data
- Annually:
- ✓ Full portfolio analysis with counterfactuals
 - ✓ Review fairness implications of counterfactual scenarios
 - ✓ Update customer communication strategy

COMPLIANCE NOTES:
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- ✓ Transparency: Counterfactuals provide CLEAR explanation
 - ✓ Fairness: Same counterfactual factors apply to all groups
 - ✓ Explainability: Addresses "why not approved" directly
 - ✓ Documentation: Save counterfactuals for audit trail
- ⚠ Caution: Ensure counterfactuals are FEASIBLE for applicants to avoid frustration or legal challenge